Ethics and Information Technology (2025) 27:18
https://doi.org/10.1007/s10676-025-09825-6

ORIGINAL PAPER

# Trust and power in Airbnb's digital rating and reputation system

**Tim Christiaens**[1]

Accepted: 25 February 2025 / Published online: 27 March 2025
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**Abstract**
Customer ratings and reviews are playing a key role in the contemporary platform economy. To establish trust among strangers without having to directly monitor platform users themselves, companies ask people to evaluate each other. Firms like Uber, Deliveroo, or Airbnb construct digital reputation scores by combining these consumer data with their own information from the algorithmic surveillance of workers. Trustworthy behavior is subsequently rewarded with a good reputation score and higher potential earnings, while untrustworthy behavior can be algorithmically penalized. However, can these technologies themselves be trusted? Are they reliable communicators of trustworthy market behavior and do they fairly distribute the benefits and burdens of the online economy among platform companies, consumers, and workers? The implementation of Digital Reputation Systems (DRSs) has obvious advantages, like reducing adjudication costs and enhancing consumer trust, but there are also disadvantages that are predominantly offloaded on workers. Taking Airbnb's DRS as my case study, I use Stephan Lessenich's theory of the externalization society to argue that DRSs concentrate the benefits of the platform economy among consumers and platform companies while pushing negative externalities mostly unto workers.

**Keywords** Trustworthy Technology · Trust ethics · Digital rating systems · Airbnb · Algorithmic Surveillance

*Why did you wish me milder? Would you have me*
*false to my nature? Rather say I play the man I am.*
—William Shakespeare, *Coriolanus.*

You would not get into a stranger's car, but you regularly take an Uber; you would not let a stranger into your home, but you look for a dogsitter on Pawshake; you would never sleep on a stranger's couch, yet your next city trip destination will be an apartment you booked on Airbnb. The online platform economy encourages individuals across the world to intimately engage with strangers, but that requires securing trust among platform users first. Especially in the platform-mediated gig economy, companies often use algorithmic surveillance to identify and deactivate unreliable workers and reward trustworthy behavior (Crawford, 2021; Delfanti, 2021; Lamers et al., 2022; Levy, 2023; Wood et al., 2019). However, they also gather a substantial amount of data from customer ratings and reviews. 'Surveillant consumers' (Stark & Levy, 2018) conduct part of the supervision process on platform companies' behalf to ascertain the trustworthiness of workers. Platform companies subsequently combine this information from ratings and reviews with their own data about worker performance to establish 'digital reputation systems' (DRSs) that act as trust-building devices. Digital reputation scores facilitate online interactions by informing platform users about each other's trustworthiness. However, can the trust-building technologies themselves be trusted? Do customer ratings and reviews provide a representative picture of gig workers' trustworthiness, or do they merely present a distorted viewpoint on customers' own biases? And even if DRSs were fully reliable, can all stakeholders trust that digital platforms design their reputation scores with everyone's interests in mind? After all, platform companies are profit-oriented businesses that strive to maximize the number of transactions conducted on their platforms from which they generate revenue. They might be more interested in the quantity of transactions rather than their quality or the wellbeing of all parties involved.

In this paper, I aim to study whether we should "trust our selves" (Kiran & Verbeek, 2010) to DRSs with Airbnb's DRS as my case study. New digital technologies are not mere passive instruments for human use but mediate social relationships in ways that transform human subjectivity itself (Bergen & Verbeek, 2021; Danaher & Sætra, 2022; Verbeek,

✉ Tim Christiaens
t.christiaens@tilburguniversity.edu

[1] Department of Philosophy, Tilburg University, Warandelaan 2, 5000LE Tilburg, Netherlands

2006). Once people rate and review each other's social persona on the Airbnb platform, human beings adapt their behavior to the platform's reward-and-punish structure. We should hence investigate whether the digital infrastructure underlying DRSs can be trusted. That requires assessing not only whether DRSs are reliable tools for measuring human social qualities but also whether they align with our normative expectations (Nickel, 2011, 355; Ferrario et al., 2020, 530).[1] Kiran and Verbeek (2010, 424) propose two criteria to evaluate technologies' trustworthiness. Users must possess the ability to, firstly, understand how these technologies affect their lifeworld, and secondly, to autonomously influence this process to fit their desires and needs. This paper contributes to the first task: we must understand how the transformations DRSs on platforms like Airbnb normatively affect human users. After sketching the context of Airbnb's DRS, I argue that Airbnb's DRS fails to merit our trust by being both unreliable and unfair in how it distributes the benefits and burdens of online social interactions. I use Stephan Lessenich's theory of the externalization society to argue that Airbnb's DRS concentrates the benefits of platformized tourism among the company itself and consumers, while offloading disadvantages disproportionately to Airbnb hosts. According to Lessenich, contemporary capitalist societies tend to reproduce socio-economic and political inequalities by concentrating the benefits of economic activities in the hands of corporations and their consumers, while offloading the adverse side-effects of economic development, so-called 'negative externalities', to vulnerable others. While corporations establish contracts with their consumers to mutual benefit, hidden costs are externalized unto third-party others, who have little say over the contract's terms yet bear its consequences. I argue that DRSs are the platform economy's equivalent of an externalization machine. The benefits of platformized transactions are primarily allocated to platform companies and their consumers. Online ratings offer a cheap and easy format for worker surveillance, which encourages consumers to trust platforms like Airbnb, while it maximizes data creation and revenues for platform companies at minimal costs. The social and psychological costs, however, are predominantly borne by platform workers.

[1] In this paper, I will stick to a minimal definition of technological trust. For more advanced discussions and the debate about whether 'trust' can apply to technologies at all, I refer the reader to (Coeckelbergh 2012; Keymolen and Van Der Hof 2019; Nickel 2022; Ryan 2020; Taddeo 2009; Taddeo and Floridi 2011).

## Airbnb and its hosts

After an Airbnb stay, tourists receive an automated message asking them to rate their host on 8 metrics: overall experience, cleanliness, accuracy, check-in, communication skills, location, value, and amenities. They also write a brief review for the host's profile page. Contrary to most other digital platforms, Airbnb establishes a bidirectional review process that also asks hosts to review their guests. However, bidirectionality does not imply symmetry: hosts can only respond to guests' reviews or write a brief review of their own. Host-written reviews often have little effect, since Airbnb is predominantly interested in monitoring hosts' service provision. Its DRS measures the latter by combining customer ratings with data about, for example, the speed with which hosts respond to messages or their acceptance rate for bookings (Roelofsen & Minca, 2018, 176). This combined score ultimately determines hosts' visibility on Airbnb's platform. As we will see, this design choice can create asymmetrical power-relations between hosts and guests. Hosts' income and livelihood depend on their ranking in Airbnb's search engine, whereas guests only need to fear the minor chance that hosts reject their application after bad reviews. That rarely happens, since hosts depend on bookings for income and Airbnb encourages hosts to automatically accept reservations via its 'Instant Booking'-system. Moreover, rejected tourists can still opt for a hotel or another Airbnb host, but downgraded hosts receive almost no booking requests if their lodging is too low on Airbnb's suggestion list. When conflicts between guests and hosts do occur, Airbnb also often sides with guests, since it regards guests as its primary audience (Gainsforth, 2019, 183).

But are Airbnb hosts workers at all, let alone gig workers? The average Airbnb host is wealthier and more privileged than the stereotypical gig worker in ride-hailing or food delivery (Ravenelle, 2019, 40; Schor et al., 2020, 843; Muldoon, 2022, 16). At the very least, they can afford housing with an extra bedroom, which suggests Airbnb hosts do not suffer from the same precarious living conditions as other gig workers. Airbnb also grants some freedoms to hosts that gig workers at Uber or Deliveroo lack, like the freedom to set their own prices or build personal clientele. A large section of Airbnb hosts consequently consists of professional real-estate agencies, who buy multiple apartments in touristic cities to put them on Airbnb. This group has more in common with corporate landlords than impoverished gig workers. A central ethical debate surrounding this type of Airbnb hosting concerns not data-driven surveillance but urban gentrification: touristic cities become unlivable if short-term tourism occupies too much rental space (Bosma & van Doorn, 2022; Cocola-Gant & Gago, 2021; Gainsforth, 2019; Spangler, 2020). Most of the problems I discuss in this

paper do not apply to this group of professionalized investor-hosts but to the still considerable group of small-scale amateur hosts who rent out parts of their own living space. In contrast to large-scale real-estate investors, amateur hosts genuinely depend on good ratings to pay their own rents. As research in other sectors of the platformized gig economy has already demonstrated, moral problems usually appear most clearly among those individuals who depend on their platform-supported income for everyday survival (Anwar & Graham, 2021; Rosenblat, 2018; Schor et al., 2020).

Small-scale Airbnb hosts are an exemplary case study for trust ethics for two reasons. Firstly, establishing trust between hosts and guests is crucial when they share intimate living spaces for extended stretches of time. A Deliveroo courier only meets customers briefly at the door, so consumer feedback is not the most important assessment metric. Deliveroo's DRS hence privileges other data, like average speed, to measure performance. Among Airbnb hosts, however, emotional labor is a crucial part of the gig (Knaus, 2020, 72). Customer ratings are subsequently not mere reflections of an impersonal service but of hosts' abilities to project likeability. If hosts want to get ahead, they *must* score well on customer ratings and reviews. Personalized service and emotional involvement are their unique selling point vis-à-vis corporate landlords. Secondly, Airbnb guests themselves choose their lodgings directly on the Airbnb platform, and they explicitly rely on ratings and reviews. On Uber, consumers just book a ride, and the algorithm unilaterally determines to which driver it links the request. When consumers give a bad rating, the driver's digital reputation score drops and Uber might deactivate these workers, but good ratings have a limited effect on drivers' income. Consumers have no say in who is selected, so ratings and reviews have little influence over the allocation of work. Airbnb's platform, on the other hand, only offers consumers a list of suggestions, but it does not make the choice of lodging for them. This makes the DRS not only a means for punishing bad behavior, but also for rewarding good service. Consumers rely heavily on ratings and reviews, so it is crucial not just to avoid a bad digital reputation but also to cultivate a good reputation. Hosts want to be among the top suggestions in their area to attract the attention of as many consumers as possible. They hence work hard not just to avoid deactivation but also to merit visibility on the platform and convince consumers inspecting their profiles.

## The advantages of DRSs: avoiding adjudication overload and enhancing trust

To understand the advantages of Airbnb's DRS, it is helpful to know why digital ratings were invented in the first place. AuctionWeb–known today as eBay–was the first to introduce online consumer ratings on its platform (Etzioni, 2019, 6; Tadelis, 2016, 322). The company was founded in 1995 by computer engineer Pierre Omidyar as an online marketplace where strangers could meet and exchange goods. However, the management team was quickly overwhelmed by complaints from duped platform users (Tarnoff, 2022, 73). Protected by the relative anonymity of the internet, ill-intentioned individuals sold goods at substandard quality or misrepresented their merchandise. AuctionWeb consequently suffered from adjudication overload: there were too many complaints for too few employees. The company risked obtaining significant extra labor costs to manage these customer complaints. It introduced online ratings to cheaply outsource quality control and risk assessment to users themselves. AuctionWeb enabled individual users to assess each other's trustworthiness and communicate this to others directly, which would pre-emptively resolve the risks of trading among strangers. Ratings functioned as a token of past trustworthiness and a predictor of future trustworthiness. They were a disciplining force on bad traders, while rewarding good traders for their efforts. A pleasant side-effect was that platform users would automatically gravitate toward high-reputation individuals and avoid low-reputation individuals, reducing the overall number of complaints. Airbnb and other platform companies have subsequently copied AuctionWeb's solution.

Underlying the challenge of adjudication overload is a problem of market trust: online transactions among strangers often involve information asymmetries that disadvantage ignorant trading partners. In the case of Airbnb, hosts can easily lie about their lodgings or post misleading pictures on their profile, while guests pay in advance for their stay (Etzioni, 2019, 4). Such information asymmetries foster distrust among trading partners (Schor & Attwood-Charles, 2017, 5–6; Ravenelle, 2019, 29; Yang et al., 2019, 1980). In the words of George Akerlof (1970), digital platforms resemble 'markets for lemons', i.e. a market for second-hand cars where consumers lack sufficient information to detect untrustworthy marketing (Ert & Fleischer, 2019, 280). While it is relatively easy to tamper with second-hand cars, ignorant consumers are hardly equipped to understand when they are being duped. They thus bear potentially high 'costs of dishonesty' and will hence think twice before engaging in market transactions (Akerlof, 1970, 495). Free markets require trust, but the latter requires proof that signals trading partners' trustworthiness. If such markers of trust are lacking, risk-averse individuals avoid entering uncertain markets, which reduces the overall number of transactions and consumer welfare. In the case of Airbnb, potential tourists hesitate to book lodgings on Airbnb's platform fearing that hosts are giving them misleading information about substandard living spaces. This hurts not only consumer welfare, but also well-intentioned hosts, who receive fewer reservations

for their lodgings, and Airbnb. The latter earns revenue by gathering data and charging fees on every transaction, but if the number of transactions declines, so will revenues and the amount of data.

Solving the 'market for lemons'-problem requires trust-building devices that amend the information asymmetry on the consumer side, according to Akerlof (1970, 499). These mostly take the form of surveillance systems (Whelan, 2019). Akerlof mentions, for instance, the possibility of licensing sellers or introducing third-party experts to perform quality checks. Airbnb, for example, authenticates user IDs to check whether Airbnb users have a criminal record (Bridges & Vásquez, 2018, 2058). It also performs AI-driven trust scoring to predict and prevent potentially fraudulent transactions (Etzioni, 2019, 5). However, Airbnb's platform facilitates millions of bookings every year across the globe. Hence why platform companies normally opt for Akerlof's second solution: reputational feedback. By publicizing the reputation of sellers, consumers amend their ignorance thanks to information from peers who have previously interacted with the seller. Hosts subsequently engage in practices of self-branding to enhance their digital reputations and thereby attract more guests (Knaus, 2020, 76; Yang et al., 2019, 204). Just like companies protect their brand reputation to retain customers, individual hosts build a personal brand out of ratings and reviews to secure future business opportunities. This process starts with uploading a personal photo and editing the profile in ways that signal a welcoming and reliable atmosphere (Ert & Fleischer, 2019, 282; Ye et al., 2017, 2), but ratings and reviews form the bedrock of hosts' reputational persona. Ratings and reviews compose the first impression potential guests have when inspecting hosts' profile pages. The latter function as a 'constantly updated CV' (Zhu et al., 2019, 3703) that guests gauge before they pick their lodging. DRSs thereby visualize reputation in ways that encourage consumers to put trust in hosts.

DRSs have clear benefits insofar as they diminish adjudication costs for platform companies and build market trust. Its primary beneficiaries are consumers, who can rely on ratings and reviews to amend information asymmetries. The platform company itself also benefits because more market trust entails more transactions and hence higher revenues (Srnicek, 2017, 45). With more consumers turning to Airbnb, the company expands its database and the revenues coming from commission fees. Hosts also profit indirectly because the increased market trust means that more consumers are willing to spend money on the platform. Trustworthy hosts are, furthermore, rewarded for their good conduct with higher positions on Airbnb's suggestion lists and can back up their promises with evidence. However, this leaves one question unresolved: can we trust the technology itself? The economic perspective on market trust focuses on the establishment of trusting relationships between market agents, but it thereby ignores the impact of the technological mediators it prescribes. Akerlof and others champion trust-building devices like reputation scoring technology, but they imagine these technologies as transparent communicators of people's trustworthiness. Research from ethics of technology on trust suggests, however, that these technologies themselves must also meet particular standards to merit our trust (Coeckelbergh, 2012; Etzioni, 2019; Keymolen & Van Der Hof, 2019). Technologies transform the human subjectivities and relationships they mediate, and hence we must assess whether we agree with the kind of lifeworld these technologies foster. Social media, for example, changes how people conduct political deliberations. It renders the political arena more accessible to voices otherwise marginalized (Benkler, 2006; Schneider, 2022), but it also boosts the influence of inflammatory and polarizing opinions (Davies, 2021; Gerbaudo, 2018; Sunstein, 2018). Social media algorithms introduce a 'new structural transformation of the public sphere' (Habermas, 2023), just as radio and television have done in the past. Ethical and political thought must hence reflect on whether these algorithms align with our views on good political deliberation.

To merit our trust, DRSs must first be reliable, i.e. they must adequately perform the function for which they were designed. If DRSs are created for measuring the trustworthiness of market agents, then they must faithfully reflect people's performance. However, as we will see, many imperfections and distortions can creep in during the translation of human interactions into datafied reputation scoring. Secondly, technologies must align with our normative expectations concerning the distribution of the benefits and burdens of online market transactions. Mere reliability is an insufficient marker for trustworthiness because destructive technologies can still reliably perform their function when they harm their victims. Trustworthy technologies, in other words, must equitably consider all stakeholders affected in their implementation. In what follows, I suggest that this is currently not the case in Airbnb's DRS insofar as it tends to concentrate benefits among consumers and the platform itself while offloading disadvantages unto hosts.

## The externalization society and DRSs: What if DRSs are (un)reliable?

In this section, I argue that Airbnb's DRS enhances trust among platform users at low adjudication costs, but there are also negative side-effects to continuous peer-to-peer assessment that are disproportionately borne by Airbnb hosts. Briefly put, DRSs produce benefits primarily for platform companies and consumers–and indirectly for platform workers–but also negative externalities are almost exclusively concentrated on the side of hosts. The latter bear

most downsides of having consumers continuously rate and review service workers' actions. I will first introduce Lessenich's theory of the externalization society to conceptualize this imbalance. Afterwards, I apply Lessenich's approach to Airbnb's DRS, focusing first on the negative externalities arising from the DRS' failure to reliably measure hosts' performance accurately. DRSs are designed to measure platform users' trustworthiness as market agents, but they regularly fail to the reliability standard, which often offends certain normative expectations as well. Airbnb's ratings, for example, can present guests' racially biased opinions about their hosts as fact, which not only renders them unreliable predictors of hosts' actual performance but also devices for racial discrimination. Secondly, Airbnb's DRS also generates negative externalities predominantly affecting hosts when their metrics are working reliably. Even if DRSs adequately represent hosts' trustworthiness as market agents, for instance, they might still impose constant customer surveillance on hosts. Since the latter's income and future opportunities depend on continuous judgment from guests, such invasive surveillance refashions human social interactions into a dystopia of '*Black Mirror*'-like all-pervasive surveillance without escape.

## The externalization society and platform capitalism

In economics, negative externalities are costs that contracting parties in a transaction offload on non-involved others. Lessenich generalizes this framing into a sociological interpretative scheme. In his view, the production of economic prosperity among the well-off is often intertwined with hidden mechanisms that displace negative side-effects toward vulnerable others. Lessenich's *Living well at Others' Expense* focuses on international environmental policy, where the rich Global North often offsets the costs of its own energy-intensive mass consumerist lifestyles on impoverished regions in the Global South. But Lessenich explicitly presents externalization as a general explanation for other social inequalities as well, like migration or the sexual division of labor in the household (Lessenich, 2018, 2019, 92–126; 2020). What these diverse cases have in common is a pattern of cost-shifting in which corporations and their consumers use the dependency of workers and others to unburden themselves of particular risks. Lessenich (2019, 1–8), for instance, discusses the case of a mining disaster near the Brazilian Rio Doce from 2015. A transnational corporation, Samarco, intensified mining operations to please consumer demand for metal resources in the Global North. But while Samarco made profits and consumers received access to cheap metals, the local population in Brazil was exposed to the risks of intensified mining. In 2015, a dam holding toxic wastewater burst, flooding the entire area. Samarco tried to avoid paying damages by blaming the incident on an earthquake, but eventually the Brazilian state forced Samarco to bear responsibility for the negative externalities it produced.

Digital platforms exhibit similar externalization dynamics, as Lessenich (2019, 139–140) illustrates with social media content moderation. Social media prohibit offensive content like hate speech, child pornography, and graphic violence, but the filtering work resists complete automation and is hence regularly performed by low-paid clickworkers (Gillespie, 2018, 101–102; Scholz, 2017, 21). These are often young, un(der)employed individuals who urgently need income (Jones, 2021, 53). To optimize the social media experience, the costs of sifting through online garbage is, in other words, externalized unto precarious clickworkers. Just like Samarco exploited the dependency on mining jobs of Brazilians to cut costs on environmental management, social media companies cut costs on content moderation by outsourcing clickwork to impoverished workers. These costs do not only concern time and effort, but also a high psychological toll (Gillespie, 2018, 123). Content moderators often leave their jobs after a few months and subsequently suffer for years from sleeping disorders, depression, or PTSD-like symptoms. While social media users enjoy social connection, and social media companies profit from selling targeted advertisement space, the negative externalities of connecting the global population are dumped on invisibilized workers.

These divergent cases exhibit a common logic of externalization, which combines the incorporation of benefits into a privileged core with the outsourcing of collateral damage to the periphery. "In the externalization society, power consists of the opportunity for transferring the costs for one's way of life to others—and this opportunity is distributed in a structurally unequal fashion" (Lessenich, 2019, 42). Economic inequalities are, in this optic, effects of asymmetrical power-relations. The poor depend on the privileged core for their sustenance, and the latter institutionalize these dependencies into seemingly neutral contractual relations. Histories of power ultimately determine who is in the position to offload the negative externalities of their own lifestyles unto others, while hoarding benefits for themselves. Since the Brazilian miners need jobs, they can be negotiated into bearing the environmental risks of industrialized mining; since the un(der)employed youth need to pay rent, social media companies can displace the psychological costs of offensive online content. The unequal nature of these exchanges is hidden underneath a veneer of contractual egalitarianism. This is not necessarily a matter of evil intentions, though. Individuals are often stuck in social logics that push them to reproduce logics of externalization despite their best intentions. In global environmental policy, for instance, governments in the Global North depend for their legitimacy and national stability on the ability to stimulate economic growth and consumer demand (Lessenich, 2018, 425). Failure to

procure these goods results in lost elections or populist protests, which disincentivizes governments from imposing the negative side-effects of energy-intensive lifestyles unto their own citizens.

The same constraints apply to for-profit platform companies, whose financial viability depends on continued growth. Platform companies like Airbnb, Uber, or Spotify heavily rely on investors' funds to keep themselves financially afloat until they acquire market leadership (Christophers, 2020, 221; Van Doorn & Badger, 2020, 1480). In the meantime, these companies grow their data assets and try to monopolize the social spheres in which they operate (Christiaens 2022a, 74; Couldry & Mejias, 2019, 107). Their continued existence depends on acquiring an indispensable key position in society so that they can become profitable without depending on external investors anymore (Srnicek, 2017, 103). If Airbnb wishes to financially survive, it must try to monopolize the online tourist industry. Once it captures a sufficiently large share of online transactions in tourism, it can build a valuable database of guests' and hosts' personal data, and demand a larger cut from each individual transaction. Airbnb and other platform companies can temporarily afford to run losses because they rely on investor money, but when they fail to establish market dominance, investor funds will inevitably run out. Companies like Airbnb, Uber, and Spotify, which are not yet profitable, are still in the gray zone concerning their financial long-term viability ([Author]Christiaens 2022b, 53). Growth through the attraction of more consumers is hence essential to today's platform businesses.

To support these growth ambitions, companies often make their platforms freely accessible and promise users comfortable convenience and cheap services in exchange for data (Sadowski, 2020, 57). In the case of Uber, the platform allows consumers to circumvent expensive taxi companies to directly connect with independent drivers, while Airbnb cuts out hotels as unnecessary middlemen from the tourists' holiday planning. According to Floridi (2016, 188), "this is the sense in which a [platform] technology (especially an interface) is transparent: not because it is not there, but because it delivers its services so efficiently, effectively, and reliably that its presence is imperceptible". Airbnb claims to democratize global travel by facilitating tourists to directly connect with locals without the mediation of hotels. In exchange for a convenient travel experience, Airbnb receives a small fee and data about people's travels. This bargain offers cheap comfort for consumers, but there is little regard for the consequences borne by third parties not involved in that transaction. Hosts and locals are responsible for recreating the welcoming tourist experience Airbnb promises. Residents of touristic city centers dominated by Airbnb bear the burden of increased rental prices and unlivable, 'Disneyfied' neighborhoods (Spangler, 2020). They are the victims of 'predatory inclusion': they are included into Airbnb's business model yet only as dumping grounds for the negative externalities of mass tourism (Tarnoff, 2022, 132). In general, labor platforms like Airbnb, Uber or Deliveroo are infamous for offloading costs and risks of their business unto workers as well (Ravenelle, 2019, 36; Srnicek, 2017, 83). They, for instance, prefer to classify workers as independent contractors rather than employees because this legally transfers the risks of illness, accidents, non-productive downtime, etc., to workers.

In the remainder of this article, I highlight how Lessenich's approach discloses Airbnb's DRS as an externalization machine. Though it creates trust and a comfortable consumer experience with limited adjudication costs, it offloads the costs of maintaining the tourist experience unto hosts and partly, as we just witnessed, local neighborhoods. Again, dependency and power-relations underlie this imbalanced distribution of costs and benefits. If potential hosts wish to make money through Airbnb, they sign the terms and conditions and agree to unilaterally set requirements written to maximize consumer satisfaction with little regard for hosts' own well-being (Scholz, 2017, 75). There is no possibility to negotiate which aspects of their work are measured, how the different metrics are weighed, what personal boundaries hosts can set. This unequal predicament forms the background for the logic of externalization embodied in DRSs. Given that platform companies' business model tends toward monopolization, hosts often have few exit options in the form of competitor rental platforms if they do not support Airbnb's terms of agreement. They might try other platforms, like Booking.com, but these have similar business strategies as well as analogous terms and conditions and DRS. If people wish to monetize their lodgings as short-term rentals, they must probably accept some version of the same agreement, whether through Airbnb or another platform company. I distinguish between two kinds of externalities: some problems emerge from the fall-out of a DRS' failure to reliably measure hosts' performance, while a second collection of problems relates to the scenario in which DRSs succeed at reliably measuring work performance.

## Problems from DRS unreliability

Airbnb's DRS is sometimes untrustworthy because it fails to reliably represent platform users' actual trustworthiness as market agents. By allowing for (a) consumer biases, (b) algorithmic biases, and (c) algorithmic errors, DRSs create unfair market outcomes that disproportionately affect platform workers, who oftentimes depend for their livelihood on online platforms and their DRSs. In these three scenarios, platform users should not trust DRS technology because it fails the reliability standard of technological trust.

### Consumer bias

Airbnb's DRS condenses many ratings and reviews into an easily understandable metric of 1 to 5 stars with some brief comments attached. However, this quantitative presentation gives an 'aura of unassailable objectivity' to essentially subjective judgments (Mau, 2019, 160). Consumers browsing Airbnb's platform often put a lot of unwarranted trust in the DRS. The subjective nature of guests' experiences entails that they might have been using wildly different standards to judge hosts' performance (Knaus, 2020, 78; Roelofsen, 2022, 108). The DRS artificially renders them equivalent through its quantitative metric, but this offers no guarantees regarding accuracy. Airbnb users overestimate each other's ability to equivalently and fairly rate hosts' performance on a scale from 1 to 5. DRSs are only as good as the data they receive as input, but data from subjective judgment is often riddled with unfair assessments or implicit biases. Some guests are, for instance, unrealistically demanding and cast vindictive judgments when disappointed. They also sometimes exaggerate these frustrations to get refunds from Airbnb. Roelofsen (2022, 108), for example, writes about guests photographing dust on the furniture to demand refunds, while Gainsforth (2019, 183) reports about guests writing misleading reviews about syringes scattered in the bathroom to get refunds. Hosts can try to respond to these abusive reviews, but the costs of setting the record straight are exclusively for hosts, with slim chances of success. Usually, they respond to the review but fail to remove the negative review on their public profile. Abusive reviews can thereby permanently hurt a business with no cost for the lying consumer.

Apart from intentionally unfair judgment, there is also the matter of implicit bias in guests' ratings (Ravenelle, 2019, 35–36; Tan et al., 2021, 7). Hosts of a minoritarian race or class are sometimes evaluated quite differently on metrics like communication skills or cleanliness. Black hosts in New York, on average, must ask 12% less than non-black hosts for lodgings of similar quality (Edelman & Luca, 2014).[2] Aided by the illusory objectivity of DRSs, guests' prejudices negatively affect the earning potential of black hosts. Even the mere presence of hosts' personal photos in Airbnb's search results dampens the financial prospects of black hosts (Ert & Fleischer, 2019; Ye et al., 2017). Consumer ratings and reviews should hence not be regarded as objective performance measurements but as "opinions embedded into mathematics" (O'Neil, 2017, 21). If DRSs assess the quality of hosts' services, then consumer data is a highly biased input sample skewing Airbnb's algorithmic assessment of hosts' work. Hosts alone, however, bear the costs of these inaccuracies.

### Algorithmic bias

Apart from users' biases, platform design and algorithmic decision-making also produce biases of their own (Amoore, 2020, 74–75). Hiring algorithms, for instance, have created controversy not just because they translate biased training data into discriminatory hiring decisions, but also because the machine-learning software of these algorithms sometimes invents new discriminatory standards. Hiring algorithms unfairly sidelining female applicants are well-known (Criado-Perez, 2020), but other algorithms have discriminated on entirely new and unforeseen grounds, like rejecting applicants because of the web browser they use (Wachter, 2022). Not just a DRS' content matters, but also its form (O'Neil, 2017, 115). The design of assessment algorithms like DRS not only makes use of biased data-inputs, but can also bias outcomes themselves through its design choices. Platform design determines what data users are likely to produce, which behaviors are legible as 'datafiable' input, or how this data is handled and translated into an algorithmic decision. The criteria Airbnb's DRS lists guide guests to a particular way of seeing that focuses attention on particular hoisting qualities, like the presence of amenities or the speed of the check-in process (Roelofsen, 2022, 100).

Admittedly, a full analysis of algorithmic bias in Airbnb's DRS is impossible without behind-the-scenes insight into its patent-protected software. Here, I stick to the weaker claim that DRS design influences consumer bias through its suggestions for what to rate and review. Technological design might look neutral, but it often shields political choices from scrutiny behind a façade of convenience (Couldry & Mejias, 2019, 23; Srnicek, 2017, 47). Airbnb's search results, for instance, used to immediately show hosts' profile pictures in the search engine. When the company discovered that this design choice encouraged racial bias among potential guests, it removed profile pictures from the search screen (Ert & Fleischer, 2019, 283). Another design feature with crucial effects on digital reputations is Airbnb's choice to de-anonymize reviews (Bridges & Vásquez, 2018; Meijerink & Schoenmakers, 2021; Teubner & Glaser, 2018; Zervas et al., 2021). Users post reviews in their own name, which has created a positivity bias in ratings. Most hosts have a J-shaped rating graph, entailing that almost all ratings tend toward 5 stars. Such abundance of positivity might seem good, but appearances are deceiving. Hosts' visibility on Airbnb's search engine lists does not depend on the absolute numbers of digital reputation scores but on their relative position vis-à-vis competitors. Having a 4.8 score means nothing if others have 4.9. Such high scores subsequently

[2] Airbnb hosts themselves also display racial bias, with black guests' reservations being 16% more likely to face rejection, despite Airbnb's in-built incentives for automatic acceptances (Edelman et al., 2017).

increase the risk that one negative review upsets hosts' entire business (Ravenelle, 2019, 13). The platform design, however, offloads such risks only unto hosts.

### Algorithmic error

Apart from systematic biases in the data or platform design, there is, lastly, the matter of algorithmic mistakes (Amoore, 2020, 118; Pasquale, 2016, 40). Software bugs also produce negative externalities for hosts, as the latter were not involved in making the DRS yet unilaterally bear the costs of its failings. The burden is on hosts to discover glitches, report them, and demand reparation. When these issues occur systematically, one starts suspecting platform companies use 'bugs' strategically to hoard more income for themselves at the cost of unknowing workers (Benjamin, 2019, 80). Rosenblat (2018, 114–128), for instance, suspects Uber of strategically inserting 'mistakes' in its payment algorithm to not pay out tips or cancellation fees to drivers. Grocery-delivery platform Instacart also had a 'glitch' that systematically withheld tips from workers and only fixed its mistake when nation-wide protests broke out (Dickey, 2019). For Airbnb, the Airbnb Plus program offers an example of negative externalities from algorithmic error (Gainsforth, 2019, 151–52). It is a quality label for hosts with highly exclusive lodgings that allows them to charge more and appear higher in the search results. Members need a good digital reputation and conform to pre-established hospitality standards, like the presence of certain amenities. They must prove their compliance with photographic evidence to enter the program, but Airbnb does not check these pictures manually. It outsources these assessments to AI-driven image recognition software that often makes mistakes, like failing to recognize water bottles or misidentifying shadows as carpet stains. Applicants subsequently receive non-sensical feedback which they must obey to acquire the Airbnb Plus status. Hosts' costs of reporting mistakes and the time lost on e-mailing helpdesk workers are not recompensed. Moreover, these requests for rectifying algorithmic errors often fail, since Airbnb's helpdesk workers are simply badly paid gig workers themselves, unable to alter the software.

## Problems from DRS reliability

The aforementioned problems justify distrusting Airbnb's DRS on the basis of its unreliability in measuring hosts' performance. However, this leaves us with the question whether we should trust DRSs in more idealized scenarios where these technologies would be able to reliably represent platform workers' performance. In this section, I argue that even in this scenario, negative externalities are offloaded unto Airbnb hosts. While the problems of (a) lateral surveillance, (b) alienation, and (c) depoliticized meritocracy do not affect the reliability of DRSs, they still offend users' normative expectations insofar as DRSs foster an unfair distribution of the costs and benefits associated with online reputation scoring.

### All-invasive lateral surveillance

If hosting constitutes a form of labor, then DRSs extend workplace surveillance further and deeper than any previous form of management supervision (Christiaens 2022b, 65). DRSs implement lateral surveillance systems, i.e. monitoring practices that empower consumers and workers to directly inspect each other in service of the platform company (Roelofsen & Minca, 2018, 176). While no traditional employee would ever allow their employer to constantly inspect their home, through the eyes of the consumer, Airbnb gets access to the private living space of its hosts. The home itself and one's intimate personality become objects of workplace surveillance (Jarrett, 2022, 93). In the context of social media, lateral surveillance threatens to undermine individuals' informational privacy and subject them to a tyranny of online publics (Lanzing, 2016, 13–14; Weissman, 2019, 33). The impact of consumer surveillance at Airbnb is similar: outsourcing this job to consumers comes with significant stress since it is often unclear to hosts what service is required to merit 5 stars in the eyes of individual guests. Hosts subsequently put up with a lot of bad behavior from guests in fear of rating retaliation (Gainsforth, 2019, 183; Ravenelle, 2019, 45). Since any interaction with the guest carries the risk of a downgrade and hence loss of future income opportunities, hosts feel forced to compromise on their personal boundaries. They let offenses slide that in other human interactions, encounters in which homely intimacy is real rather than enforced through DRSs, would merit reprimanding.

By introducing lateral surveillance into the relation between guests and hosts, Airbnb undermines its own promise of spontaneous social intimacy and homely interaction (Knaus, 2020, 69; Jarrett, 2022, 134). Human spontaneity is not a design feature that can be planned or enforced, yet DRSs make hosts financially dependent on staged spontaneity. Simultaneously, hosts must protect their own home from excessive surveillance by stating clear boundaries for guests. The costs of maintaining this balancing act between fostering likeability and enforcing privacy are outsourced to hosts. This entails a multitude of delicate, small acts of boundary-setting with each new guest, like putting up signs asking guests to wash their own dishes, agreeing curfew times, or determining who can use the bathroom and when (Roelofsen, 2018, 25). Each of these decisions is a renewed occasion for hosts to become targets of rating retaliation. Since guests act simultaneously as customers and supervisors, they can invade hosts' private lives with minimal accountability.

Not only the private living space but also hosts' own personalities become objects of scrutiny. Airbnb's DRS asks guests to rate characteristics like their hosts' communication skills, which de facto means that hosts must adapt their personalities to fit consumer demand. In the words of Erving Goffman's theory of the social self, hosts continuously must perform the social role of a welcoming host in their own home (Knaus, 2020, 77; Roelofsen, 2018, 28). As already mentioned, this problem only applies to hosts who invite guests into their personal homes and not to professional real-estate agencies that happen to use Airbnb. Amateur hosts play a part in a script written by Airbnb's DRS, and whenever there are guests around, hosts are 'on-stage'. They subsequently learn to anticipate the objectifying gaze of the other and pre-emptively adjust their conduct to please their audience. They clean the house excessively (Knaus, 2020, 79) or put on a façade of niceness because they know guests will judge them on these metrics afterwards. Yet this has profound effects on hosts' subjectivity. According to Alison Hearn, "these conditions of work suggest a perpetually malleable, cynical and changeable style of selfhood" (426). One host subsequently confesses to Sarah Gainsforth: "I feel like a stranger in my own home" (2019, 189). It is difficult to develop a minimal self of core personal traits if one's value is constantly put in the eye of the beholder (Lasch, 2018, 44–45). Individual personalities are rendered adaptable to whatever consumers desire, which denies hosts the off-stage time to protect their minimal self (Couldry & Mejias, 2019, 156; Pasquale, 2020, 77). Permanently exposed to the judgment of others, hosts lack opportunities to 'just be themselves'. It becomes hard to preserve the integrity of the self if the value of one's personality is a quantity determined through constant peer-to-peer assessment. While the home has traditionally been the space for off-stage conduct, which allowed individuals to develop a sense of self independent from their social roles in public life, Airbnb's DRS invades this space to ensure quality service for its consumers. It fosters a 'loss of home' that comes at significant human cost for hosts (Jarrett, 2022, 152).

### Alienation

The optimal functioning of DRSs would not only indicate a loss of free, off-stage time but also an undermining of human autonomy. In Marxist literature, this problem appears under the rubric of 'alienation' (O'Regan & Choe, 2017; Glavin et al., 2021; Christiaens 2022b; Pasquinelli, 2023). While non-alienated labor puts the worker in charge of the labor process, alienated labor arises when capital takes possession over the means of production and subjects the worker's will to the command of capital (Negri, 1991, 76). According to Marx, workers in the industrial factory are alienated insofar as "an alien power, the power of the machine itself" determines the movements and thoughts of individual workers (Marx, 2005, 693). Machines are, for Marx, no longer mere means of labor but instruments through which the capitalist class subjugates workers and pushes them to work harder and longer in the service of capital accumulation. With every new commodity workers produce, they feed the 'general intellect', i.e. the technological apparatus that capital deploys to absorb workers' knowhow about how to conduct their job and unilaterally coordinate the labor process (Marx, 2005, 706).

A similarly tragic cycle is at play in DRSs: workers produce data with every gig they perform, while platform companies hoard this data and use it to perfect their fine-grained coordination of workers' conducts. Airbnb, in other words, amasses a general intellect of digital information to monitor and discipline workers' performances. Hosts subsequently experience their digital reputation not as the product of their own labor but as an 'alien power' from the past, dead labor that haunts their future earning potential and over which they have little control. Workers' agency is reduced to mindlessly enacting the will of the algorithms that manage their labor with minimal opportunities to influence the criteria that judge their performance (Christiaens 2024, 5–6). Airbnb hosts are subjected to the composite wills of many consumers collected and mediated by the DRS. The latter incarnates the wishes and expectations of many people and presents it as an easily understandable reputation score that acts like a single general intellect imposing its commands on individual hosts. Hosts subsequently adapt their behavior to conform to the expectations of the digital apparatus that coordinates this scoring activity. In their own hosting, workers are expected to enact the will of the general intellect rather than their own. If hosts resist these expectations, they risk a negative review and thereby a loss of income.

One ironic side-effect of DRS-induced alienation is the homogenization of services subjected to DRSs (Tan et al., 2021, 28; Mau, 2019, 138). Hosts fear falling in Airbnb's search engine results if their actions deviate from the platform's expectations. They subsequently opt for non-offensive forms of conduct, choices that minimize the risk of scaring off potential guests. Anything that could be considered off-putting or too unique is avoided in favor of the generically comfortable. As Weismann observes, "one 'optimizes' oneself online by presenting oneself as a likeable, approved member of the crowd, that has the same thoughts, feelings, interest, and behaviors as others in the crowd, and hides inclinations that might prove otherwise" (2019, 35). Since potential guests choose their lodgings via first impressions through Airbnb's search engine, distinctive self-presentation can work against hosts' business chances. Airbnb hosts thus remove personal items from their living spaces, pictures of family members or events with friends, and local cultural artifacts. They opt for a highly generic interior design style

dubbed 'the Airspace-aesthetic' (Chayka, 2016). Lodgings in many global cities look surprisingly similar, with white walls, beige leather sofas, black and white pillows, and cheap yet reasonably trendy Ikea furniture. Ironically, Airbnb promises an authentic experience with local hosts, yet its DRS incentivizes hosts to strip their lodgings of local authenticity in favor of universal mass appeal.

### Depoliticized meritocracy

DRSs structure the platform economy as winner-takes-all markets. On Airbnb's website, consumers often look exclusively at the search engine's first few results. Income opportunities for hosts are concentrated at the top based on past performance. DRSs, in other words, seem to institute meritocratic markets, in which those who have proven their worth in the past keep receiving more, while underperformers receive fewer guests at lower prices. It seems like individual merit in hosting services directly translates into financial success. Meritocracies certainly have benefits (Sandel, 2021, 33): they increase efficiency by allocating tasks to those who have a proven record of aptitude, and they fight human biases or mistakes that otherwise disadvantage those who deserve more. That is why AuctionWeb introduced ratings and reviews in the first place. Perfectly reliable DRSs measure and rank people according to merit alone. However, strong competition for so few lucrative spots on Airbnb's suggestion list comes at considerable social and psychological costs offloaded on hosts.

Moreover, "there is no neutral definition of 'merit'" (Mijs, 2016, 20). What counts as good service is a contestable standard on which Airbnb now unilaterally decides. Airbnb currently rewards a very particular kind of merit yet remains silent about the underlying choices of what constitutes good hosting services. Even if its measurements of hosts' performance would be perfectly accurate, there are still potential problems about how transparent the search algorithm should be, who owns the reputation data, how much competition is desirable, who is allowed to list their lodgings on the platform, etc. By instituting a market where income opportunities are distributed according to Airbnb's definition of merit and scoring standards, DRSs hide the political embedding of this market. Hosts are expected to accept the short-term rental market when they accept the terms and conditions, rather than question how digital reputation is measured and how it impacts hosts' opportunities. Side-lining collective agency and self-government in favor of a technocratic order in which the meritorious govern without contestation is a general characteristic of meritocracy (Sandel, 2021, 108; Young, 1994, 38). Airbnb risks creating an equivalent system in which data experts unilaterally design DRSs, while hosts must accept their lot without questioning. The workforce is not treated as a collective subject but is fragmented into a disconnected series of hyperindividualized entrepreneurs of their own lodgings (Mau, 2019, 168). These individuals are constantly in competition with each other rather than collectively organizing to defend their shared interests vis-à-vis the data experts of Airbnb. The latter remain in the background yet determine the standards for who can make a living on Airbnb and who cannot.

There is also a psychological cost to meritocratic government insofar as even top performers are never safe from downgrading and hence keep pushing themselves to improve their business until they burn out (Sandel, 2021, 178; Young, 1994, 80–81). On Airbnb's platform, it is not the absolute numbers of ratings that count but the relative value of one's digital reputation compared to others. This leads to a meritocratic arms race, in which every improvement for one participant triggers incentives for rivals to outperform them. Ultimately, Airbnb hosts endlessly tweak their profiles, change the interior design of their lodgings, adapt their behavior, to marginally improve their digital reputation. Since their earning potential often hinges on a 0.1 difference in ratings, any small adjustment might have grand effects on overall revenues. Hosts end up constantly editing their profiles and personalities to become a 'Super Self' (Roelofsen, 2022, 106). However, this practice can easily transform into an exhaustingly endless cycle of self-optimization (Hearn, 2010, 423; Mau, 2019, 84). There is always one more thing to improve upon. The burdens of the stress coming from permanent precarity and feelings of inadequacy are one-sidedly borne by hosts, while consumers and Airbnb profit from the endlessly improving services hosts offer. The losers, those who fail to entertain the arms race for self-optimization, are quickly demoralized and leave the platform (Mijs, 2016, 22).

## Conclusion

Online markets in the platform economy encounter serious challenges concerning market trust. In the case of Airbnb, it requires considerable effort to convince tourists of paying for a room in a stranger's lodging. To cultivate market trust and diminish the load of adjudicating consumer complaints, companies like Airbnb have designed digital rating and review systems that allow platform users to assess each other's trustworthiness directly. This benefits not only consumers, who now possess more information to evaluate workers' future performance, but also platform companies, who generate revenue from increased online transactions, and platform workers themselves, who profit from more demand for their services. DRSs have, in other words, quickly become an indispensable technology for the infrastructure of the platform economy. However, this leaves the question of whether DRSs themselves can be trusted to mediate between platform users. Technological

mediation transforms human social interactions, and we should assess in advance whether these transformations are desirable and fair. Technologies can be trusted to facilitate human sociality if and only if they reliably enact the function they were designed for and avoid offending our normative expectations. However, Airbnb's DRS currently does not meet these standards. It acts as an externalization machine that concentrates the benefits of increased market trust and online transactions among consumers and platform companies themselves, while offloading the negative externalities from these transactions predominantly unto workers. These negative externalities come in two varieties. Firstly, DRSs are sometimes unreliable tools for the measurement and communication of platform workers' performance. In the case of Airbnb, they regularly consolidate consumer biases, algorithmic biases, and algorithmic errors. Secondly, even if DRSs would reliably measure platform workers' performance, they would still act as externalization machines by offloading the costs of lateral surveillance, alienation, and depoliticized meritocracy unto platform workers.

**Acknowledgements** I would like to thank the referees for their helpful comments, as well as the participants at the Future of Work conference of ETUI in 2023 and Paolo Santori and Willem Van Der Deijl-Kloeg for their advice on previous versions of the manuscript.

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